Usability Testing Report: Software Engineering

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1 Revision History

Date	Developer	Notes
March 29th	Team #16, The Chill Guys	Initial scaffold: sections, methodology shell, placeholders for post-study results.
March 29th	Swesan Pathmanathan	UT-2/UT-3/UT-2R task text aligned to app UI; trip methodology and results/recommendation shells for trip owner.
March 29th	Swesan Pathmanathan	Methodology: two Android devices, Render API, Neon DB, account-deletion reset; UT-2R out of scope; illustrative $N = 10$ simulated results and discussion filled.

2 Symbols, Abbreviations and Acronyms

symbol	**description**
HCI	Human–Computer Interaction
SUS	System Usability Scale
CSUQ	Computer System Usability Questionnaire (optional instrument)
UT	Usability test; task identifiers `UT-1`, `UT-2`, ...
FR	Functional Requirement (from SRS)
NFR	Nonfunctional Requirement (from SRS)
SRS	Software Requirements Specification
API	Application Programming Interface
T	Task identifier in session scripts (e.g., T1, T2)

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This document describes a usability evaluation plan and **illustrative** quantitative and qualitative results for Hitchly, a campus-focused rideshare application for the McMaster community. Task identifiers `UT-`n denote scenarios in Section 6. Results and discussion reflect **simulated persona walkthroughs** ($N = 10$), as stated in Sections 5 and 9.

3 Introduction

3.1 Purpose and Scope

This report documents a **structured, quantified** usability-style evaluation of Hitchly, aligned with HCI reporting practice. It specifies the **goals** of testing, the **tasks** performed, the **questions** and **survey instruments** used for feedback, and the **metrics** used to summarize outcomes. Section 5 and Section 9 state clearly that aggregates are **illustrative** ($N = 10$ simulated personas) rather than a live external-participant study.

The evaluation targets the mobile experience (and any supporting flows such as email verification) for McMaster-affiliated riders and drivers. It complements verification activities in the Verification and Validation (VnV) plan by focusing on learnability, efficiency, error tolerance, and satisfaction for representative users rather than on code-level correctness alone.

3.2 Relation to Requirements

Usability objectives align with non-functional expectations in the SRS for understandable, low-friction flows (e.g., sign-up, offering or requesting a ride, confirming a match, payment-related steps where applicable). Findings may be traced to specific FR/NFR identifiers when reporting issues or recommendations.

3.3 Document Structure

Section 4 states test goals. Section 5 describes participants, setting, and procedure. Section 6 lists task scenarios. Section 7 covers moderator questions and surveys. Section 8 defines quantitative measures and analysis. Sections 9 and 10 contain **example** results and recommendations from simulated walkthroughs. Appendix 10.3 holds questionnaire text.

4 Goals of Usability Testing

The following goals guide session design and success criteria. They should be measurable using the metrics in Section 8.

1. **Onboarding and trust:** Assess whether participants can complete registration, McMaster-affiliated verification, and role selection (rider/driver) without undue confusion or abandonment.
2. **Core rider journey:** Evaluate discoverability and completion of finding or requesting a ride, understanding trip details, and confirming participation within the app.
3. **Core driver journey:** Evaluate offering a ride, setting or understanding route and schedule-related inputs, and managing passenger-related actions as implemented.
4. **Matching and trip state:** Observe comprehension of match suggestions, trip status, and navigation between related screens.
5. **Error recovery:** Record where users deviate, hesitate, or commit errors, and whether they recover without moderator intervention (within a reasonable threshold).
6. **Subjective satisfaction:** Quantify perceived ease of use and confidence via standardized scales (e.g., SUS) and structured debrief responses.

Team members may refine goal wording after piloting the script with one internal participant to ensure tasks remain feasible on the build under test.

5 Methodology

5.1 Study Design

Evaluation followed the **task scenarios** in Section 6 using **moderated, think-aloud style walkthroughs**. Due to course timeline constraints, sessions used **simulated personas** (team-assigned roles and scripted reactions) rather than external recruitment. Section 9 reports **illustrative aggregates** ($N = 10$) consistent with those walkthroughs; they demonstrate the reporting format and plausible friction points but **do not constitute a live human-subjects study** or statistically validated benchmark.

5.2 Participants

- **Simulated sample:** $N = 10$ notional participants (personas P1–P10), each mapped to rider- or driver-oriented task subsets as in Section 9.
- **Role mix:** Six sessions emphasized **rider** tasks (UT-2), four emphasized **driver** tasks (UT-3); all personas walked through UT-1, UT-4, and UT-5 where applicable.
- **Scope: UT-2R** (recurring rides) was **out of scope** for this evaluation; drivers used **ONE-TIME** only in UT-3.

5.3 Environment and Materials

- **Devices:** Two **Android** phones running the Hitchly Expo client against the same backend.
- **API and database:** Application traffic went to the **Render**-hosted API; persistent data used **Neon** (PostgreSQL).
- **Network:** Stable Wi-Fi; same `EXPO_PUBLIC_API_URL` for both devices (see `apps/mobile/lib/trpc.ts`).
- **State reset between scenarios/personas:** Test **accounts were deleted** and flows restarted **from fresh registration** so trips, requests, and matches did not carry over spuriously.
- **Ethics note:** No external participants were recorded; simulated quotes in Section 9 are **paraphrased narrative examples** from team walkthroughs, not verbatim recordings.

5.3.1 Trip flows (configuration used)

Recurring rides (UT-2R): Not evaluated; documented only for future reference in Section 6.
Accounts: New `@mcmaster.ca` test accounts created per reset; email verification exercised as in UT-1; home address set before UT-2/UT-3 when testing posting or search.

5.4 Session Outline

1. Briefing and simulated consent script (Appendix 10.3).
2. Optional background questionnaire (not all personas recorded).
3. Fixed order: UT-1 → UT-2 or UT-3 (by persona) → UT-4 → UT-5.
4. Post-task prompts from Section 7 (ease rating captured per task).
5. SUS-style ratings assigned per persona for aggregation (Appendix 10.3).
6. Team debrief and log update before account deletion / next persona.

5.5 Data Recording

For each task, record: start/end time, success or failure, number and type of assists (minimal prompt vs. direct help), and brief qualitative notes (quotes, confusion points). Use a session log template (spreadsheet or form) referenced in Section 9.

6 Task Scenarios

Each scenario below maps to goals in Section 4. Labels and navigation for UT-2 and UT-3 reflect the current Expo app (`apps/mobile`) as of this document. Success criteria are **binary** unless otherwise noted. **UT-2R was out of scope** for the evaluation reported here (see Section 5).

6.1 Scenario UT-1: First-time access and verification

- **Context:** You are a new McMaster student who has just installed Hitchly.
- **Task:** Create an account with a McMaster-affiliated email, complete any verification step required, and reach the main home or dashboard screen.
- **Success criteria:** Account created; email verification completed (or flow correctly attempted per test account setup); lands on post-auth home without moderator rescue beyond one standard hint.

6.2 Scenario UT-2: Rider—find or request a ride

Prerequisite: Profile must include a default location (home address); otherwise the app shows **LOCATION REQUIRED** and **GO TO PROFILE**. Moderator either confirms the test account is pre-configured or assigns a short sub-task to add an address.

- **Starting point:** Rider home with the **FIND A RIDE** form visible. If the participant is on **My Rides**, tap **Find a Ride** first.
- **Task:**
  1. On **FIND A RIDE**, set **ROUTE** direction (to/from McMaster) using the in-form control.
  2. If travelling **to** campus, set campus **drop-off** for the moderator's landmark (modal lists McMaster drop-off options).
  3. Set **date** and **desired arrival** time per moderator script.
  4. Tap **SEARCH FOR RIDES**.
  5. Wait for **SEARCHING...** to finish.
  6. If matches exist: read **SWIPE TO MATCH**. **Swipe right** to **request** a ride; **swipe left** to pass (per moderator).
  7. Optionally tap a card to open **Trip Details**.
  8. If no matches: note **NO MATCHES FOUND**. Participant may tap **NEW SEARCH** (valid path if empty-state usability is documented).
- **Success criteria (binary):**
  - Participant completes **SEARCH FOR RIDES** from a valid form state.
  - They reach one of: swipe deck ($\geq$ 1 card); **NO MATCHES FOUND**; or **Trip Details** from a card tap.
  - No persistent validation errors.
  - They can state that swiping right requests a ride.
- **Partial credit (optional):** If the study records ordinal difficulty, note hints needed for drop-off or direction.

6.3 Scenario UT-2R: Recurring ride (reference only—not evaluated)

Not in scope for this study. The following remains as documentation for future testing. In the reported sessions, drivers always selected **ONE-TIME** in UT-3.

- **Driver path (future):** Open **Post Ride**; under **REPEAT**, choose **RECURRING**; set direction, **DEPARTURE**, **AVAILABLE SEATS**; **PUBLISH RIDE**.
- **Rider path (future):** From **My Rides**, open a recurring booking; use **REQUEST NEXT** when shown.

6.4 Scenario UT-3: Driver—offer a ride

- **Starting point:** Driver **My Trips**; tap **Post Ride** (opens modal titled **Post a Ride**).
- **Prerequisite:** Profile must list a home address; otherwise the form shows a callout to set it in Profile. Moderator uses a seeded driver account with address filled, or allocates time for profile setup.
- **Task:**
  1. Under **DIRECTION**, choose **TO MCMASTER** or **FROM MCMASTER** as specified.
  2. Confirm **ROUTE** shows home and McMaster (read-only); under **REPEAT**, choose **ONE-TIME** (recurring was out of scope).
  3. Set **DEPARTURE** (datetime) and **AVAILABLE SEATS** (stepper).
  4. Tap **PUBLISH RIDE**.
- **Success criteria (binary):** Submission succeeds (loading completes, modal closes or trip appears in **My Trips**); no blocking validation errors. Participant sees the new trip in the driver's trip list or receives a clear, understandable error (document if failure is a usability defect).

6.5 Scenario UT-4: Matching or trip follow-up

- **Context:** A compatible ride or match is available (moderator may set state in advance).
- **Task:** Accept, decline, or confirm the match/trip as appropriate to the prototype state.
- **Success criteria:** Participant completes the intended state transition or explains blockage clearly; note any UI ambiguity.

6.6 Scenario UT-5: Profile or settings (optional)

- **Task:** Update a non-sensitive profile field or locate notification/payment settings.
- **Success criteria:** Participant finds the correct area within a time threshold (set in Section 8) or documents failure.

Table 1: Task scenario summary ($N = 10$ simulated walkthroughs; UT-2R not run).

ID	Focus	Notes
UT-1	Onboarding / verification	9/10 success
UT-2	Rider search / swipe / request	7/10 success
UT-2R	Recurring ride	Not in scope
UT-3	Driver post ride	8/10 success
UT-4	Match / trip state	8/10 success
UT-5	Profile / settings	9/10 success

7 Session Questions and Survey Instruments

7.1 Pre-session Background Questions

Q1: How often do you commute to or from McMaster (or campus)?

Q2: Do you usually drive, ride as a passenger, or use transit?

Q3: Have you used any rideshare or carpool apps before? Which?

Q4: What device are you using today?

7.2 Post-task Probing Questions

After each scenario (Section 6), ask as appropriate:

PT1: What did you think that screen was asking you to do?

PT2: Was anything surprising or unclear?

PT3: On a scale of 1 (very difficult) to 7 (very easy), how easy was that task? (Record numeric response.)

PT4: Did you feel you could trust the information the app showed (e.g., cost, other user, timing)?

7.3 Post-session Interview Questions

PS1: Overall, how would you describe your experience using Hitchly today?

PS2: What would you change first if you could?

PS3: Would you use this for real commuting if it were available? Why or why not?

PS4: Any concerns about safety, privacy, or payments?

7.4 Standardized Questionnaires

- **System Usability Scale (SUS):** Administer the standard 10-item SUS immediately after tasks; compute the standard 0–100 score per participant. Full wording appears in Appendix 10.3.
- **Optional supplementary Likert blocks:** e.g., trust in campus verification, clarity of pricing—use 5- or 7-point scales with anchors "Strongly disagree" / "Strongly agree"; include items in Appendix 10.3.

All numeric survey responses feed the quantitative analysis described in Section 8.

8 Metrics and Quantification

This section defines how usability is measured to satisfy **rigorous, quantified** reporting. Fill aggregate values in Section 9 after data collection.

8.1 Task Performance

- **Task success rate:** For each UT-n, proportion of participants who meet success criteria without failure. Report per-task and overall.
- **Time on task:** Elapsed time from moderator prompt to success or abandonment; report median, interquartile range, and outliers.
- **Assists:** Count moderator prompts (minimal hint vs. substantive help); define weighting if a composite "error score" is used.
- **Critical incidents:** Count usability problems by severity (blocks task / major / minor); use a simple rubric agreed before sessions.

8.2 Subjective Measures

- **SUS score:** Per participant and mean $\pm$ standard deviation across N.
- **Post-task ease (1–7):** Mean per task from PT3 (Section 7).
- **Optional Likert composites:** Mean scores per construct (e.g., trust, clarity).

8.3 Analysis

- Summarize quantitative results in tables; use figures (e.g., bar chart of success rates) if required by the course.
- Thematic analysis on open-ended responses: tag recurring themes and link to screenshots or timestamps when recordings exist.
- Map top issues to SRS FR/NFR or design modules where applicable for traceability.

8.4 Thresholds (optional targets)

Document any targets set before testing (e.g., at least 80% success on UT-1, median SUS $\geq$ 68). Revise after pilot if unrealistic for the current build.

9 Results

9.1 Limitations of reported data

Aggregates below are **illustrative**: ten **simulated persona walkthroughs** on two Android devices (Section 5). Metrics, times, and quotations are **constructed for course reporting** to reflect plausible usability issues observed during structured team runs—not a sample of independent external users.

9.2 Participant Summary

- **Notional sample size:** $N = 10$ personas (P1–P10).
- **Role mix:** Six personas prioritized **rider** flows (UT-2); four prioritized **driver** flows (UT-3); all completed UT-1, UT-4, and UT-5 where applicable.
- **Environment:** Two **Android** phones; API on **Render**; database on **Neon**; state reset by **deleting accounts** and re-onboarding between personas.
- **Out of scope: UT-2R** (recurring) was not executed.

9.3 Quantitative Summary

9.4 Trip-related tasks (UT-2, UT-3)

UT-2 (rider search / swipe): Seven of ten personas completed without substantive help. Failure modes (simulated): two stalled on campus **drop-off** selection (unclear landmark naming); one exited to **NO MATCHES FOUND** and did not retry within the session timer. Median time from opening **FIND A RIDE** to tapping **SEARCH FOR RIDES**: 48 s (IQR 38–62 s). Median time from search results to first intentional swipe: 41 s.

Table 2: Task success ($N = 10$ simulated sessions).

Task ID	**Successes**	**Rate (%)**
UT-1	9/10	90
UT-2	7/10	70
UT-3	8/10	80
UT-4	8/10	80
UT-5	9/10	90

Table 3: Time on task, seconds (median and IQR).

Task ID	**Median**	**IQR**
UT-1	95	72–128
UT-2	152	118–198
UT-3	88	71–112
UT-4	74	58–96
UT-5	42	33–58

Table 4: Moderator assists (minimal hint vs. substantive help) across 10 personas.

Assist type	**Count**	**Per persona (mean)**
Minimal hint	14	1.4
Substantive help	3	0.3

Table 5: SUS scores (illustrative per-persona ratings, 0–100 scale).

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
64	70	68	76	82	71	73	69	75	78

Mean = 72.6; standard deviation = 5.4; min = 64; max = 82.

Table 6: Post-task ease (1 = very difficult, 7 = very easy), mean across personas.

After task	Mean ease
UT-1	5.9
UT-2	5.1
UT-3	5.6
UT-4	5.7
UT-5	6.0

UT-3 (driver post ride): Eight of ten succeeded on first **PUBLISH RIDE** attempt. Two required a hint about the home-address prerequisite callout. Median time from **Post Ride** to successful publish: 72 s (IQR 59–94 s).

UT-2R: Not evaluated (out of scope).

9.5 Qualitative Observations

Themes (illustrative): (1) campus **drop-off** labels require cognitive mapping to real buildings; (2) **TO MCMASTER / FROM MCMASTER** is understood after one use but causes hesitation on first exposure; (3) swipe affordances are learnable once **SWIPE TO MATCH** is read; (4) driver **ONE-TIME** vs. recurring control was unused but visible—no confusion reported for this study.

Representative paraphrased reactions (simulated, not verbatim recordings):

- "I see a long list of drop-off names—I am not sure which one is closest to my class." (UT-2)
- "I thought TO MCMASTER meant something about my ride direction, not my commute intent—took a second to trust it." (UT-2)
- "After the address warning, Publish Ride was straightforward." (UT-3)
- "Swipe right to request makes sense after I read the footer text." (UT-2)

Critical incidents (counts, illustrative): three **major** tied to UT-2 drop-off/modal navigation; eight **minor** (wording, back navigation, empty match state).

9.6 Screenshots or Figures

No figures included: evaluation used internal walkthroughs without participant media release.

10 Discussion and Recommendations

10.1 Interpretation

Illustrative results align with the goals in Section 4: **onboarding** (UT-1) and **profile-adjacent** tasks (UT-5) scored highest ease; **rider discovery** (UT-2) showed the lowest success rate and longest times, mainly around **drop-off selection** and first-time **direction** semantics. **Driver posting** (UT-3) was generally efficient once the home-address prerequisite was satisfied, supporting the cost-sharing value proposition but surfacing **gating copy** as a friction point. **Matching follow-up** (UT-4) was moderate—consistent with dependency on prior trip state and match availability.

Trade-offs: Tighter campus landmark fidelity (trust, clarity) increases choice complexity; swipe-based matching is engaging but demands clear onboarding for first-time users.

Limitations: $N = 10$ **simulated** personas; no external diversity; two Android devices only; **Render/Neon** latency not stress-tested; **UT-2R** omitted; quotes are **stylized** from team debriefs, not transcripts.

10.2 Recommendations

10.2.1 General (all task areas)

10.2.2 Trip, search, and post-ride (Swesan)

10.3 Traceability

High-priority items map to SRS usability and clarity goals (low-friction flows, understandable trip and match actions). Link specific issues to GitHub tick-

Issue	Evidence	Sev.	Proposed change	Status
Drop-off landmark discoverability	UT-2: 70% success; ease 5.1; simulated quote on list length	Maj.	Add search/filter or map snippet for McMaster drop-offs; group by campus zone	Open
First-time direction semantics	UT-2: hesitation theme; assists on route	Min.	Short inline tooltip for TO/FROM McMaster	Open
Empty match recovery	UT-2: one persona did not retry NEW SEARCH	Min.	Prompt after empty state: suggested time adjustment	Open
Address prerequisite visibility	UT-3: 2/10 hints; UT-3 ease 5.6	Min.	Link callout to one-tap Profile prefill when empty	Open

Table 7: Cross-cutting usability recommendations (illustrative evidence).

Issue	Evidence	Sev.	Proposed change	Status
Search-to-swipe latency perception	UT-2 median 152 s total; SEARCHING state	Min.	Skeleton or progress text on match fetch	Open
Publish confirmation feedback	UT-3: 8/10 first-try success	Min.	Stronger success toast + auto-return to My Trips	Open
Recurring not tested	UT-2R out of scope	—	Schedule UT-2R in a future study with real users	Planned

Table 8: Trip-related recommendations (UT-2, UT-3).

ets when implemented (e.g. drop-off UX, empty-state copy, profile shortcut from post-ride callout).

Appendix — Survey Instruments and Consent

Consent Script (Outline)

- Purpose of the session; voluntary participation; right to stop.
- Recording (screen/audio) if used; storage and who may view data.
- No penalty for withdrawal; approximate duration.
- Contact for questions.

System Usability Scale (SUS)

For each item, use a 5-point scale from **Strongly disagree** (1) to **Strongly agree** (5). Score using the standard SUS formula (odd items: score = response -1; even items: score = $5-$ response; sum $\times 2.5$ for 0–100).

1. I think that I would like to use this system frequently.
2. I found the system unnecessarily complex.
3. I thought the system was easy to use.
4. I think that I would need the support of a technical person to be able to use this system.
5. I found the various functions in this system were well integrated.
6. I thought there was too much inconsistency in this system.
7. I would imagine that most people would learn to use this system very quickly.
8. I found the system very cumbersome to use.
9. I felt very confident using the system.
10. I needed to learn a lot of things before I could get going with this system.

Optional Likert Items (Trust and Clarity)

Use 7-point Likert scales (1 = Strongly disagree, 7 = Strongly agree). Adapt wording after pilot if needed.

1. I trust that other users on this app are affiliated with McMaster as claimed.
2. The app's pricing or cost information was clear.
3. I understood what would happen after I confirmed a ride or match.
4. I felt my personal information was handled appropriately for a student project prototype.

Session Log Template (Fields)

- Participant ID (anonymous); date; moderator; device/OS; build.
- Per task: start time, end time, success Y/N, assists, notes.
- SUS total; optional Likert means.